NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 26/03/2026
Time: 3 Hrs | GTM – 05 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	A	B	A	B	C	A	B	C	C
11	12	13	14	15	16	17	18	19	20
C	C	B	C	A	C	A	B	B	A
21	22	23	24	25					
18	4	51	2	400					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	A	B	D	D	A	D	C	A	A
36	37	38	39	40	41	42	43	44	45
B	C	D	C	A	C	A	B	C	B
46	47	48	49	50					
1143	222	2	6	101					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	D	D	C	B	C	C	B	A	C
61	62	63	64	65	66	67	68	69	70
C	B	D	C	C	A	A	B	A	D
71	72	73	74	75					
4	2454	14	1	3					

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HINTS
PHYSICS

1. We know that, $\frac{E_0}{B_0} = C$

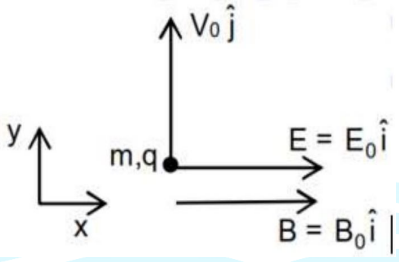
Speed of light in vacuum

Amplitude of electric field, $E_0 = B_0 C = 3\times10^{-8}\times3\times10^{8} = 9N/C$ Only option A matches.

When an electromagnetic wave passes through a medium electric field vector, magnetic field vector and direction of propagation of the wave are in mutually perpendicular directions.

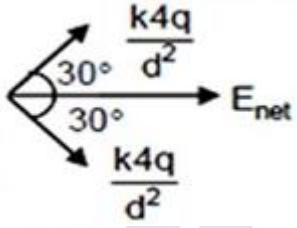
$E = E_0 \sin[kx + \omega t]$

2. As $\vec{V} = v_0\hat{j}$ (magnitude of velocity does not change in y - x plane)



$(2v_0)^2 = v_0^2 + v_x^2;\ v_x = \sqrt{3}v_0$ $\quad\therefore \sqrt{3}v_0 = 0 + \frac{qE}{m}t; t = \frac{\sqrt{3}mv_0}{qE_0}$

3.



Electric field due to B and C is along OC and

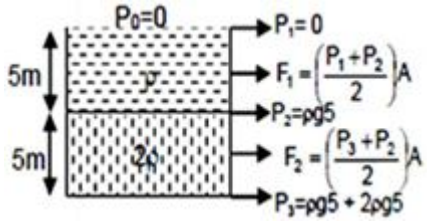
Electric field due to A is along OA

As per figure, the net horizontal component of electric field is

$E_{net} = \frac{4kq}{d^2}\times 2\cos 30^0 = \frac{q\sqrt{3}}{\pi\varepsilon_0 d^2}$

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4.



Force on upper half $= P_{avg}A = \frac{P_1+P_2}{2}A = \frac{0+\rho g5}{2}A = \frac{\rho g5}{2}A$

Force on lower half $= P_{avg}A$

$= \frac{P_2+P_3}{2}A = \frac{(\rho g5)+(\rho g5+2\rho g5)}{2}A = 2\rho g5A \therefore \frac{F_U}{F_L} = \frac{\rho g5A}{2(2\rho g5)A} = \frac{1}{4}$

5. Applying Momentum Conservation,

$$m_1\vec{u}_1 + m_2\vec{u}_2 = (m_1+m_2)\vec{V}_c \Rightarrow m(u\,\hat{i}) + mu\left(\frac{\hat{i}+\hat{j}}{2}\right) = 2m\vec{V}_c \Rightarrow \vec{V}_c = \frac{3}{4}u\,\hat{i} + \frac{1}{4}u\,\hat{j}$$

Now, magnitude of the speeds are,

$u_1 = u\ u_2 = \frac{u}{\sqrt{2}}\ V_c = \frac{\sqrt{10}}{4}u$ K.E loss $= K_i - K_f$

$$\Rightarrow \text{K.E loss} = \left(\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2\right) - \frac{1}{2}(m_1+m_2)V_c^2$$

$$= \left(\frac{1}{2}mu^2 + \frac{1}{2}m\frac{u^2}{2}\right) - \frac{1}{2}(2m)\frac{10}{16}u^2 = \frac{1}{8}mu^2$$

6. Given : $u_x = 3\ m/s, a_x = 6\ m/s^2, a_y = 4\ m/s^2$

By applying equations of Kinematics

$$y = u_yt + \frac{1}{2}a_yt^2 \Rightarrow 32 = 0 + \frac{1}{2}\times 4\times t^2$$

$\Rightarrow t^2 = 16$

$\Rightarrow t = 4\ s$

Now, $x = u_xt + \frac{1}{2}a_xt^2$

$\Rightarrow x = 3\times 4 + \frac{1}{2}\times 6\times 16$

$\Rightarrow x = 12 + 48 = 60\ m$

7. $V_A = 10\ m/s\ V_B = -20\ m/s$

Velocity of person w.r.t train $A, V_{PA} = -0.5\ m/s$

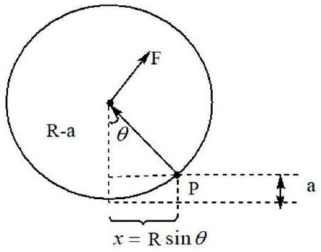
$\therefore$ Velocity of person

$\Rightarrow V_P = V_{PA} + V_A = -0.5 + 10 = 9.5\ m/s$ and

$V_{PB} = V_P - V_B = 9.5 - (-20) = 29.5\ m/s$

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8. Assume no slipping about P



Torque of F about P > torque of gravity

$\Rightarrow FR > Mgx \Rightarrow FR > MgR\sin\theta$

$\Rightarrow F_{min} = Mg\sin\theta = Mg\dfrac{x}{R}$

$= Mg\dfrac{\sqrt{R^2-(R-a)^2}}{R}$ (from figure)

$= Mg\sqrt{\dfrac{R^2-(R-a)^2}{R^2}}$

$= Mg\sqrt{1-\left(\dfrac{R-a}{R}\right)^2}$

9. Conceptual

10. Work done = change in K.E

$Pt = \dfrac{1}{2}mv^2 - \dfrac{1}{2}mv^2$ if $v = 0$

$V = \sqrt{\dfrac{2pt}{m}}$

$\dfrac{ds}{dt} = \sqrt{\dfrac{2p}{m}} \cdot t^{\frac{1}{2}}$

$\int ds = \int \sqrt{\dfrac{2p}{m}} t^{\frac{1}{2}} dt$

$S = \dfrac{2}{3}\sqrt{\dfrac{2p}{m}} t^{\frac{3}{2}}$

11. As significant figures in each measurement is 3 , the diameter should be recorded in three significant figures.

$\therefore$ key is option (3).

$B = 3\times10^{10}$

$-\dfrac{\Delta V}{V} = 0.02$

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12. $B = \dfrac{\Delta P}{-\dfrac{\Delta V}{V}} \Rightarrow \Delta P = -B\left(\dfrac{\Delta V}{V}\right)$

$= \left(3\times10^{10}\right)\left(0.02\right) = 6\times10^{8}\ \text{N/m}^2$

13. At highest point,

$T_H + mg = m\omega^2 r \Rightarrow T_H = m\omega^2 r - mg$

At lowest point,

$T_L - mg = m\omega^2 r \Rightarrow T_L = m\omega^2 r + mg$

After comparing,

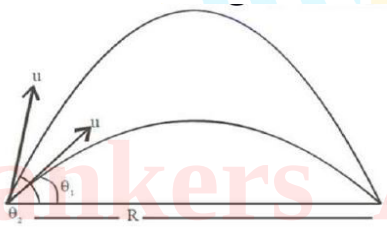
$T_H < T_L$

Thus, tension in the string is minimum at the highest position of the circular path.

14. As the sheet is very large $\hat{E}$ is independent of distance from it.

Thus $E_1 = E_2 = \sigma / 2\varepsilon_0$

15. For same range $\theta_1 + \theta_2 = 90^\circ$



$h_1 = \dfrac{u^2\sin^2\theta_1}{2g}$ $h_2 = \dfrac{u^2\sin^2\theta_2}{2g}$

$h_1 h_2 = \dfrac{u^2\sin^2\theta_1}{2g} \times \dfrac{u^2\sin^2\theta_2}{2g}$

$\theta_2 = 90 - \theta_1$

$h_1 h_2 = \dfrac{u^2\sin^2\theta_1}{2g} \cdot \dfrac{u^2\cos^2\theta_1}{2g}$

$= \left[\dfrac{u^2\sin\theta_1\cos\theta_1}{2g}\right]^2$

$= \left[\dfrac{u^2\sin\theta_1\cos\theta_1}{2g} \times \dfrac{2}{2}\right]^2 = \dfrac{R^2}{16}$

$R = 4\sqrt{h_1 h_2}$

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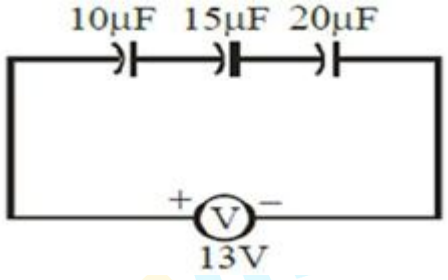
R is correct explanation of A

16. Time period of pendulum, $T_P = 2\pi\sqrt{\frac{\text{string length}}{\text{downward acceleration}}} = 2\pi\sqrt{\frac{L}{a_y}}$

If the lift is stationary, ay $= g$ Time period is $T = 2\pi\sqrt{\frac{L}{g}}$

If the lift is accelerating with $\frac{g}{6}$ Time period $T' = 2\pi\sqrt{\frac{L}{\frac{7g}{6}}} = \sqrt{\frac{6}{7}}\,T$

17.



$$\frac{1}{C_{eq}} = \frac{1}{10} + \frac{1}{15} + \frac{1}{20} = \frac{12+8+6}{120} = \frac{26}{120}$$

$$C_{eq} = \frac{60}{13}\mu F$$

$$Q = \frac{13\times 60}{13} = 60\mu C$$

Since they are in series, charge on each capacitor is same.

V_P max $= 4\ V_{wave}$

18. $$\omega A = 4\left(\frac{\omega}{k}\right) \Rightarrow A = \frac{4\lambda}{2\pi}$$

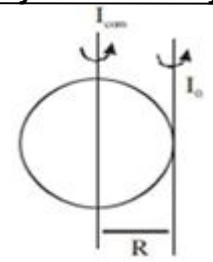
$$\lambda = \frac{2\pi A}{4} \Rightarrow \frac{20\pi}{4} \Rightarrow 5\pi$$

19. $$\lambda = \frac{kT}{\sqrt{2\pi}\,d^2P}$$

20. Solid sphere

$$I_0 = I_{com} + MR^2$$

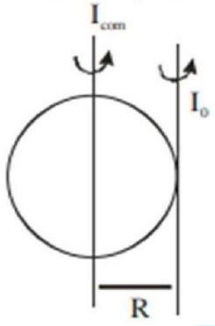
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$$I_0 = \frac{2}{5}MR^2 + MR^2$$

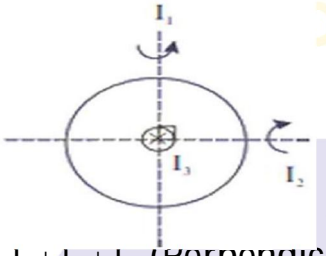
$$I_0 = \frac{7}{5}MR^2$$

Hollow sphere



$$I_0 = I_{com} + MR^2$$

$$= \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2$$



$I_1 + I_2 + I_3$ (Perpendicular axis theorem)

By symmetry

MOI about 1" and 2" Axis are same i.e.

$I_1 = I_2$

$\therefore 2I_1 = I_3 = MR^2 \left(I_{com} = MR^2\right)$

$$I_1 = \frac{MR^2}{2}$$

Similarly in disc

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$$2I_1 = \frac{MR^2}{2}\left\{I_{com} = \frac{MR^2}{2}\right\}$$

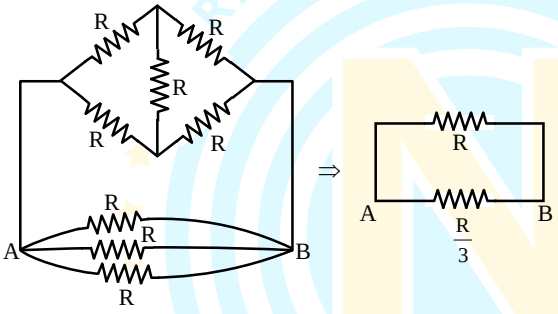
$$I_1 = \frac{MR^2}{4}$$

21. $n_1\lambda_1 = n_2\lambda_2$

$12 \times 600 = n_2 \times 400$

$n_2 = 18$ key--18

22.



$$R_{AB} = \frac{R \times \frac{R}{3}}{R + \frac{R}{3}} = \frac{R}{4}$$

23. Mass of hydrogen atom is 5times mass of colloiding particle.

Loss $= \frac{1}{2}mv^2 - \frac{1}{2}(5\ m)\left(\frac{v}{5}\right)^2$

Loss excitation energy $10.2ev = \frac{1}{2}mv^2\left(\frac{4}{5}\right)\frac{51ev}{4} = k$

$P = Vi$

$5 = 25i$

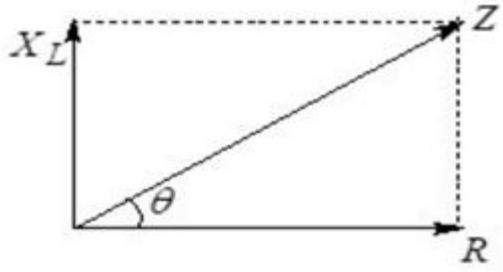
$i = \frac{1}{5}$

24. $B_y = \frac{\mu_0 NI(2r)^2}{2[(2r)^2 + d^2]^{3/2}}$

$B_x = \frac{\mu_0 NI(r)^2}{2[r^2 + (d/2)^2]^{3/2}}$ $\square$ $\frac{B_y}{B_x} = \frac{1}{2}$

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25.



$\text{Cos}\theta = 0.8 \Rightarrow \text{Tan}\theta = \frac{3}{4} = \frac{X_L}{R} \Rightarrow X_L = \frac{3R}{4}$

And $Z = \frac{5R}{4} P = \frac{V^2}{Z^2} \times R \Rightarrow \frac{P \times Z^2}{V^2} \Rightarrow R = 100\Omega$

so $X_L = 75\Omega$

For power factor to be $1, X_C = X_L$

So $\frac{1}{2\pi fC} = 75 \Rightarrow C = \frac{400}{3\pi} \mu F$

CHEMISTRY

26. $r_{2,Li^{+2}} = \frac{a_0 \times 4}{z} = \frac{a_0 \times 4}{3} = \frac{4a_0}{3}$

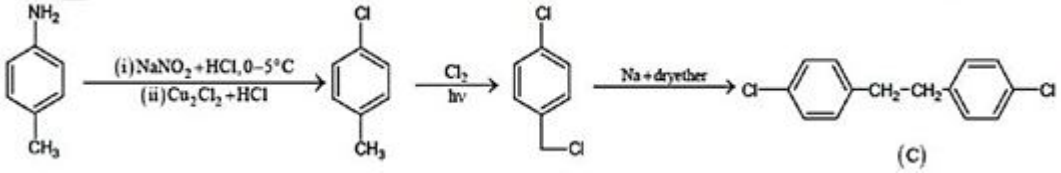
27. At equilibrium rate of forward and backward reaction becomes equal.

28. Na_2O = Basic As_2O_3 = Amphoteric
N_2O = Neutral NO = Neutral
Cl_2O_7 = Acidic

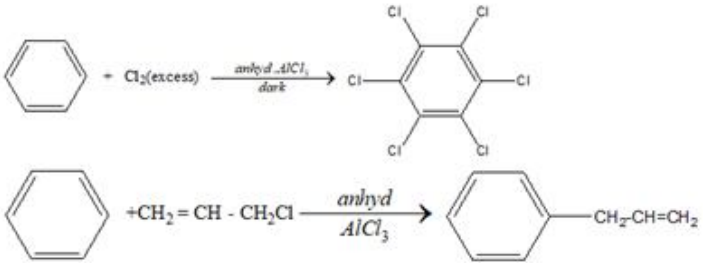
29. I^{st} I.E $\underset{(2p^3)}{N} > \underset{(2p^4)}{O} > \underset{(2s^2)}{Be} > \underset{(2p^1)}{B}$

30. Covalent radius will be smaller than atomic radius, as it involves in overlapping.

31.



32.



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33. (A) $Cr = [Ar] 3d^5 4s^1$

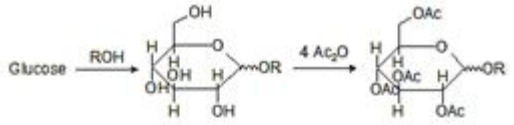
(B) $m = -1$ to $+1$

(C) According to Aufbau principle, orbitals are filled in order of their increasing energies.

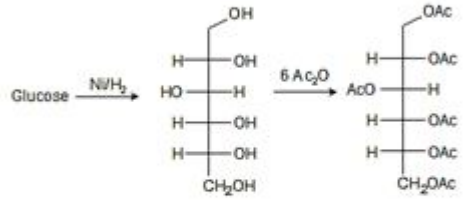
(D) Total nodes $= n-1$

34. Manganate is paramagnetic while permanganate is diamagnetic so statement I is incorrect.

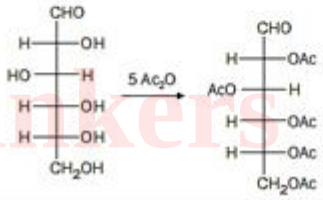
35.



$\Rightarrow x = 4$



$\Rightarrow y = 6$



36. Aliphatic amines are more basic than aromatic amines

$\therefore$ c is more basic.

' B ' is least basic since the l.p electrosn on ' N ' atom involved in aromatricity.

37. $\Delta H_{solution} = \Delta H_{rattice} + \Delta H_{hydration}$

$\Delta H_{hydration} = \Delta H_{solution} - \Delta H_{lattice}$

$= 4kJ - 788kJ$

$= -784$ kJ / mole

38. .More is the bond order, less is the bond length and vice versa.

Bond order $NO^{2+} = 2.5$

Bond order $NO^{+} = 3$

Bond order $NO = 2.5$

Bond order $NO^{-} = 2$

Bond order of NO^{-} is less, hence more is the bond length compared to other species.

39. $[HCOOH] = 0.5 ml l^{-1}$

$\Rightarrow (0.5ml \times 1.05\ gml^{-1}) HCOOH$ in 1 L

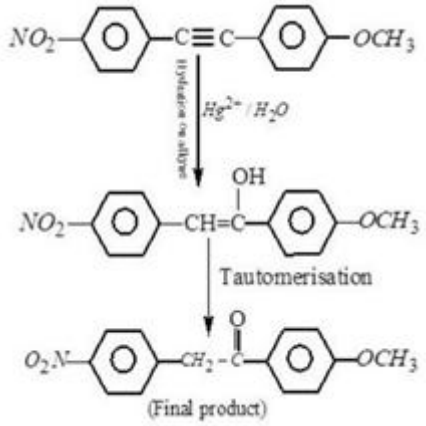
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$\Rightarrow 0.525$ gHCOOH in 1 L

$m = \dfrac{(0.525/46)}{1\text{ kg}}$ mol [Assuming a very dilute dilute solution]

$\therefore \Delta T_f = i.K_f.m \Rightarrow i = \dfrac{\Delta T_f}{k_f m} = \dfrac{0.0405\times 46}{1.86\times 0.525} = 1.9$

40.



41. $\Delta H = \sum \Delta H_C$ (Reactant) $- \sum \Delta H_C$ (Product)

$= \left[3\times(-1300) - (-3268)\right] = -632 \text{kJmol}^{-1}$

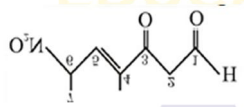
42. $Ce = Xe_{4f}{}^{1}5\,d^{1}6\,s^{2}$

$Ce^{3+} = Xe4f^{1}5\,d^{0}$

$Ce^{+4} = Xe4f^{0}5\,d^{0}$ (Noble gas configuration)

43. A is formed using OMDM reaction, whereas B is formed due to HBO reaction.

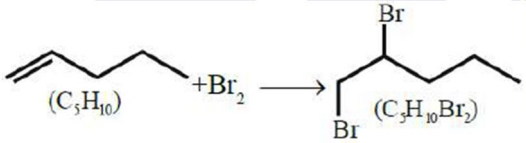
44.



4-Methyl-6-nitro-3-oxohept-4-enal

45. $BaCl_2 + SO_3^{2-} \rightarrow \underset{\text{white ppt.}}{BaSO_3}\downarrow \xrightarrow{\text{dilHCl}} \underset{\text{burning sulphur like smell}}{SO_2}\uparrow$

46.



moles of Br_2 = moles of C_5H_{10}

$\Rightarrow \dfrac{W}{160} = \dfrac{5}{70}$

$\Rightarrow W = \dfrac{5\times 160}{70}$ g

$= 11.428$ g

$= 1142.8\times 10^{-2}$ g $\approx 1143\times 10^{-2}$ g

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$$KE = \frac{hc}{\lambda} - W$$

47. $$= \frac{6.63\times10^{-34}\times3\times10^{8}}{300\times10^{-9}} - 4.41\times10^{-19}$$

$$= 6.63\times10^{-19} - 4.41\times10^{-19} = 2.22\times10^{-19}$$

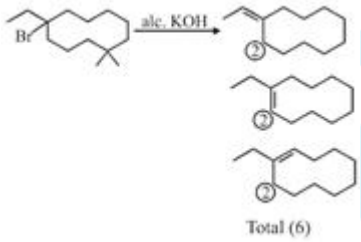
$$= 222\times10^{-21}\ \text{J}$$

$$\therefore\ \text{Ans} = 222$$

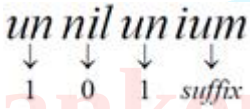
48. Number of π electron $= 2$

substitution of OH done at any position will certainly need one pi bond, hence will contain a pi bond, therefore two pi electrons.

49.



50.



MATHEMATICS

51. Given $f(a+b+1-x) = f(x)$

It implies that $f(x+1) = f(a+b-x)$

$$I = \frac{1}{(a+b)}\int_a^b x\left(f(x)+f(x+1)\right)dx \qquad ....(1)$$

$$\Rightarrow I = \frac{1}{(a+b)}\int_a^b (a+b-x)\left(f(x+1)+f(x)\right)dx \qquad ....(2)$$

Adding (1) and (2)

$$2I = \int_a^b \left(f(x)+f(x+1)\right)dx \Rightarrow 2I = \int_a^b f(a+b-x)dx + \int_a^b f(x+1)dx$$

$$\Rightarrow 2I = 2\int_a^b f(x+1)dx \Rightarrow I = \int_a^b f(x+1)dx = \int_{a+1}^{b+1} f(x)dx$$

52. $$y = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$$

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$$\Rightarrow \frac{1+y}{1-y}=\frac{8^{2x}}{8^{-2x}} \Rightarrow 8^{4x}=\left(\frac{1+y}{1-y}\right) \Rightarrow 4x=\log_8\left(\frac{1+y}{1-y}\right) \Rightarrow x=\frac{1}{4}\log_8\left(\frac{1+y}{1-y}\right)$$

$$\therefore f^{-1}(x)=\frac{1}{4}\log_8\left(\frac{1+x}{1-x}\right)=\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$$

53.Given $b|z-i|^2=|z+2i|^2$

$\Rightarrow x^2+(y-1)^2=x^2+(y+2)^2$

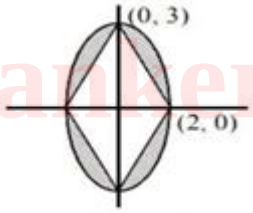
$\Rightarrow -2y+1=4y+4$

$\Rightarrow y=-\frac{1}{2}$

Given $|z|=\frac{5}{2}$

54. The area of the region outside the rhombus $\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ The area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is πab and the area of the rhombus $\frac{|x|}{p}+\frac{|y|}{q}=1$ is 2 pq Therefore, the required region is $6\pi-12=6(\pi-2)$ square units.

55.



$\Rightarrow n(B)\le n(A\cup B)\le n(U)$

$\Rightarrow 76\le 76+63-x\le 100$

$\Rightarrow -63\le -x\le -39$

$\Rightarrow 63\ge x\ge 39$

56. Let $f(x)=(\lambda^2+1)x^2-4\lambda x+2$

If $f(x)=0$ has exactly one root in $(0,1)$ then

$f(0)f(1)\le 0 \Rightarrow 2(\lambda^2+1-4\lambda+2)\le 0$

$\Rightarrow (\lambda-1)(\lambda-3)\le 0$

$\lambda\in[1,3]$

But for $\lambda=1\Rightarrow n=1$ is a repeat root

$\therefore \lambda\in(1,3]$

57 $.=\sum_{r=0}^{20} {}^{50-r}C_6 = {}^{50}C_6+{}^{49}C_6+{}^{48}C_6+\ldots..+{}^{30}C_6$

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$= {}^{50}C_6 + {}^{49}C_6 + \ldots.. + {}^{31}C_6 + \left({}^{30}C_6 + {}^{30}C_7\right) - {}^{30}C_7 = {}^{50}C_6 + {}^{49}C_6 + \ldots.. + \left({}^{31}C_6 + {}^{31}C_7\right) - {}^{30}C_7$

$= {}^{50}C_6 + {}^{50}C_7 - {}^{30}C_7 = {}^{51}C_7 - {}^{30}C_7$

58.If $x = \tan\theta$

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$$

$$= \frac{1}{2}\theta = \frac{1}{2}\tan^{-1}x$$

If $x = \sin\alpha$

$$\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right) = \tan^{-1}\left(\frac{\sin 2\alpha}{\cos 2\alpha}\right) = 2\alpha = 2\sin^{-1}x$$

Derivative of $\frac{1}{2}\tan^{-1}x$ w.r.to $2\sin^{-1}x$

$$= \frac{1}{2\left(1+x^2\right)} \times \frac{\sqrt{1-x^2}}{2} = \frac{\sqrt{3}}{10}, \text{ at } x = \frac{1}{2}$$

59. $\int \left(e^{2x} + 2e^x - e^{-x} - 1\right) e^{\left(e^x + e^{-x}\right)} dx = g(x) e^{\left(e^x + e^{-x}\right)} + c$

$$\Rightarrow \left(e^{2x} + 2e^x - e^{-x} - 1\right) e^{\left(e^x + e^{-x}\right)} = g'(x) e^{\left(e^x + e^{-x}\right)} + g(x) e^{\left(e^x + e^{-x}\right)} \left(e^x - e^{-x}\right)$$

$\Rightarrow \left(e^{2x} + 2e^x - e^{-x} - 1\right) = g'(x) + g(x)\left(e^x - e^{-x}\right)$ So, $g(x) = e^x + 1$, which gives

$g(0) = 2$

So, the correct Key is option (A)

60. $\left(\text{Cot } 27^\circ + \sin 27^\circ\right) = 1 + 2\sin 27^\circ \cos 27^\circ = 1 + \sin 54^\circ = 1 + \cos 36^\circ$

$\Rightarrow \cos 27^\circ + \sin 27^\circ = \sqrt{1 + \cos 36^\circ}$

Similarly, $\cos 27^\circ - \sin 27^\circ = \sqrt{1 - \cos 36^\circ}$ $\left(\text{since}, \cos 27^\circ > \sin 27^\circ\right)$

On subtracting, we get,

$$2\sin 27^\circ = \sqrt{1 + \cos 36^\circ} - \sqrt{1 - \cos 36^\circ}$$

$$= \sqrt{1 + \left(\frac{\sqrt{5}+1}{4}\right)} - \sqrt{1 - \left(\frac{\sqrt{5}+1}{4}\right)}$$

$$= \frac{\sqrt{5+\sqrt{5}}}{2} - \sqrt{\frac{3-\sqrt{5}}{2}}$$

$\Rightarrow 4\sin 27^\circ = \sqrt{5+\sqrt{5}} - \sqrt{3-\sqrt{5}}$

$\Rightarrow \alpha = 5 + \sqrt{5}$ and $\beta = 3 - \sqrt{5}$

$\Rightarrow \alpha + \beta = 8$

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61. Focus of $y^2=4(x-1)$ is $(2,0)$ which satisfies the equation $y=\sqrt{3}x-2\sqrt{3}$.

Hence, line $y=\sqrt{3}x-2\sqrt{3}$ is a focal chord.

Now, the length of the focal chord equals to

$4a\,\text{cosec}^2\theta$ where a=1 and $\tan\theta=\sqrt{3}\left(\text{or } \theta=60^0\right)$

$\Rightarrow$ length of chord = $4 \text{ x } \left(\text{cosec}^2 60^\circ\right)$

$=4 \text{ x } \frac{4}{3}=\frac{16}{3}$ units

62. Given,

$n_1=10,\bar{x}_1=30,CV_1=50$

$n_2=20,\bar{x}_1=25,CV_2=60$

$CV_1=\frac{\sigma_1}{x_1}\text{x } 100\Rightarrow 50=\frac{\sigma_1}{30}\text{x } 100\Rightarrow \sigma_1=15$

$CV_2=\frac{\sigma_2}{x_2}\text{x } 100\Rightarrow 60=\frac{\sigma_1}{25}\text{x } 100\Rightarrow \sigma_2=15$

Combined mean = $\frac{10 \text{ x } 30+20 \text{ x } 25}{30}=\frac{80}{3}$

$d_1=\bar{x}_1-\bar{x}=\frac{10}{3},d_2=\bar{x}-\bar{x}_2=\frac{5}{3}$

Combine variance = $\frac{n_1\sigma_1^2+n_2\sigma_2^2+n_1d_1^2+n_2d_2^2}{n_1+n_2}$

$=\frac{2250+4500+\frac{1000}{9}+\frac{500}{9}}{30}$

$=\frac{6225}{27}=\frac{2075}{9}$

63. $\frac{dy}{dx}+\frac{2^{x-y}\left(2^y-1\right)}{2^x-1}=0$

$x,y>0,y(1)=1,y(2)=\ ?$

$\frac{dy}{dx}=-\frac{2^x\left(2^y-1\right)}{2^y\left(2^x-1\right)}$

$\int\frac{2^y}{2^y-1}dy=-\int\frac{2^x}{2^x-1}dx$

$\int\frac{2^y\ln 2}{2^y-1}dy=-\int\frac{2^x\ln 2}{2^x-1}dx$

$\ln\left|2^y-1\right|=-\ln\left|2^x-1\right|+C$

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At $x=1, y=1$

64. Given that $A^T = A, B^T = -B$

(A) $C = A^4 - B^4$

$C^T = \left(A^4 - B^4\right)^T = \left(A^4\right)^T - \left(B^4\right)^T = A^4 - B^4 = C$

(B) $C = AB - BA$

$C^T = (AB - BA)^T = (AB)^T - (BA)^T$

$= B^T A^T - A^T B^T = -BA + AB = C$

(C) $C = B^5 - A^5$

$C^T = \left(B^5 - A^5\right)^T = \left(B^5\right)^T - \left(A^5\right)^T = -B^5 - A^5$

(D) $C = AB + BA$

$C^T = (AB + BA)^T = (AB)^T + (BA)^T$

$= -BA - AB = -C$

$\therefore$ Option C is not true.

65. $\lim\limits_{x\to 0} \dfrac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \dfrac{2}{3}$

$$\text{L.H.S.} = \frac{\lim\limits_{x\to 0} \alpha\left(1 + x + \frac{x^2}{2!} + \ldots\right) + \beta\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots\right) + \gamma\left(x - \frac{x^3}{3!} + \ldots\right)}{\lim\limits_{x\to 0}\left(x \cdot x^2\right) \lim\limits_{x\to 0} \frac{\sin^2 x}{x^2}}$$

$$= \lim_{x\to 0} \frac{(\alpha+\beta) + (\alpha - \beta + \gamma)x + \left(\frac{\alpha}{2} + \frac{\beta}{2}\right)x^2 + \left(\frac{\alpha}{6} - \frac{\beta}{6} - \frac{\gamma}{6}\right)x^3}{x^3 \cdot 1} + 0$$

For limit to be finite, we should have

$\alpha + \beta = 0, \alpha - \beta + \gamma = 0, \dfrac{\alpha}{2} + \dfrac{\beta}{2} = 0, \dfrac{\alpha}{6} - \dfrac{\beta}{6} - \dfrac{\gamma}{6} = \dfrac{2}{3}$

$\Rightarrow \alpha = -\beta, -\beta - \beta + \gamma = 0 \Rightarrow \gamma = 2\beta$

$\therefore \dfrac{-\beta}{6} - \dfrac{\beta}{6} - \dfrac{2\beta}{6} = \dfrac{2}{3} \Rightarrow -4\beta = 4 \Rightarrow \beta = -1$

$\therefore \boxed{\alpha = 1, \beta = -1, \gamma = -2}$

(a) $\alpha^2 + \beta^2 + \gamma^2 = 1 + 1 + 4 = 6$

(b) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = -1 + 2 - 2 + 1 = 0$

(c) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 1 - 4 - 2 + 3 = -2 \neq 0$

(d) $\alpha^2 - \beta^2 + \gamma^2 = 1 - 1 + 4 = 4$

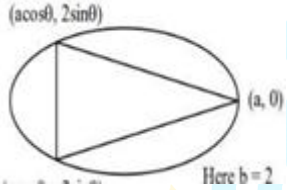
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66. $$\frac{\left(\frac{1}{p_1}\right)^2+\left(\frac{1}{p_3}\right)^2}{\left(\frac{1}{p_2}\right)^2}=\frac{\left(\frac{2\Delta}{p_1}\right)^2+\left(\frac{2\Delta}{p_3}\right)^2}{\left(\frac{2\Delta}{p_2}\right)^2}=\frac{(BC)^2+(AB)^2}{(AC)^2}$$

$\{$Where Δ represents the area of triangle ABC$\}$

$$=\frac{4+9+36+1}{16+4}=\frac{50}{20}=2.5$$

67.



$(a\cos\theta,-2\sin\theta)$

$$A=\frac{1}{2}a(1-\cos\theta)(4\sin\theta)\Rightarrow A=2a(1-\cos\theta)\sin\theta$$

$$\frac{dA}{d\theta}=2a\left(\sin^2\theta+\cos\theta-\cos^2\theta\right)\ \frac{dA}{d\theta}=0\Rightarrow 1+\cos\theta-2\cos^2\theta=0$$

$\cos\theta=1$ (Reject), $\cos\theta=\frac{-1}{2}\Rightarrow\theta=\frac{2\pi}{3}$

$$\frac{d^2A}{d\theta^2}=2a\left(2\sin^2\theta-\sin\theta\right)$$

$$\frac{d^2A}{d\theta^2}<0 \text{ for } \theta=\frac{2\pi}{3}\Rightarrow \text{ Now, } A_{max}=\frac{3\sqrt{3}}{2}a=6\sqrt{3}$$

$$\Rightarrow a=4\therefore e=\sqrt{\frac{a^2-b^2}{a^2}}=\frac{\sqrt{3}}{2}$$

68. Angle between the vectors $\vec{u},\vec{v}$ is acute

$\Rightarrow \vec{u}\cdot\vec{v}>0\Rightarrow a(\log_e b)^2-12+6a(\log_e b)>0$

$\forall b>1$

let $\log_e b=t\Rightarrow t>0$ as $b>1$

$y=at^2+6at-12$ & $y>0,\forall t>0$

$a>0, D<0\Rightarrow 36a^2+48a<0$ which is not possible

For $a<0, D<0\Rightarrow 36a^2+48a<0\Rightarrow 12a(3a+4)<0$

$\frac{-4}{3}<a<0\Rightarrow y<0$.

$\therefore a\in\phi$

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69. Let $a<b<c$ be in $A.P \Rightarrow a+c=2b$, even number

$\therefore a$ and c are both even or both odd.

The probability $=\dfrac{\binom{7}{2}+\binom{8}{2}}{\binom{15}{3}}=\dfrac{21+28}{5\times7\times13}=\dfrac{7}{65}.$

70. Any general point on the line is $(2\lambda+2,2\lambda-2,\lambda+1)$

This point most satisfy $x+y+z=21$

Hence $2\lambda+2,2\lambda-2,\lambda+1=21\Rightarrow\lambda=4$

So, the point of intersection of line and plane is (10,6, 5)

So, equation of plane P =0 is $2(x-10)+2(y-6)+1(z-5)=0$

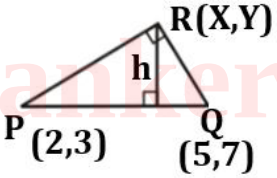
$\Rightarrow 2x+2y+z-37=0$

Hence, distance of P=0 from origin is $=\left|\dfrac{37}{3}\right|=\dfrac{37}{3}$

71. $PQ=\sqrt{3^2+4^2}=5$

Let, length of altitude through R is h

Then $\frac{1}{2}$x h x 5 =2



$\Rightarrow h=\dfrac{4}{5}$

Maximum possible length of altitude through R with right angle at R of ΔPQR (without the given area) is equal to $\dfrac{1}{2}PQ=\dfrac{5}{2}$

Now, $\dfrac{4}{5}<\dfrac{5}{2}$

So, 4 triangles are possible

72. Examination 2N,2A,2I,E,X,M,T,O

Case I All are different so ${}^8P_4=\dfrac{8!}{4!}=8\cdot7\cdot6\cdot5=1680$

Case II 2 same and 2 different so ${}^3C_1\cdot{}^7C_2\cdot\dfrac{4!}{2!}=3.21.12=756$

Case III 2 same and 2 same so ${}^3C_2\cdot\dfrac{4!}{2!\cdot2!}=3.6=18$

Total $=1680+756+18=2454$

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73. $3+(n-1)4=t_n$

$t_m=2+(m-1)7$

$t_n=4n-1$

$t_m=7m-54n-1=7m-5\ n=\dfrac{7m-4}{4}$

Put $m=-4,8$

$n=6,13 T_6=3+5(4)=23 T_{13}=3+(13-1)(4)=3+48=51$

$23+(n-1)28\le 40723+28n-28\le 40728n\le 407+5$

$n\le 14.8\ n=14$

74. $\overline{OP}=\dfrac{\lambda\bar{b}+\bar{a}}{\lambda+1}$, where $\overline{OA}=\bar{a},\overline{OB}=\bar{b}$

$SO,\overline{OB}\cdot\overline{OP}-3|\overline{OA}\times\overline{OP}|^2=6$

$\Rightarrow \bar{b}\cdot\left(\dfrac{\lambda\bar{b}+\bar{a}}{\lambda+1}\right)-3\left|\bar{a}\times\left(\dfrac{\lambda\bar{b}+\bar{a}}{\lambda+1}\right)^2\right|=6$

$\Rightarrow \dfrac{\bar{a}\cdot\bar{b}+\lambda|\bar{b}|^2}{\lambda+1}-\dfrac{3\lambda^2}{(\lambda+1)^2}|\bar{a}\times\bar{b}|^2=6$

$\dfrac{6+14\lambda}{\lambda+1}-\dfrac{3\lambda^2}{(\lambda+1)^2}\times 6=6$

$\Rightarrow \dfrac{8\lambda}{\lambda+1}=\dfrac{18\lambda^2}{(\lambda+1)^2}\Rightarrow 4(\lambda+1)=9\lambda$

$\lambda=\dfrac{4}{5}=0.80$

75. Given system of linear equations

$x+y+z=1$ ...(1)

$2x+3y+2z=1$ ...(2)

$2x+3y+(a^2-1)z=a+1$ ...(3)

Consider $a^2-1=2$

Then LHS of (2) and (3) are same but RHS are not.

Hence $a^2=3$

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